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Dynamic job allocation method of multiple agricultural machinery cooperation based on improved ant colony algorithm

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Contemporary large-scale and systematic agricultural operations demand the collaborative efforts of multiple agricultural machines with distinct functionalities. However, the failure of a single agricultural machine during collaborative operations jeopardizes the entire undertaking. To address this challenge, this paper proposes a multi-machine collaborative dynamic job allocation method based on the improved ant colony algorithm. Initially, the improved ant colony algorithm is employed to determine the optimal solution for harvester scheduling. This solution is then fed into the data conversion algorithm to acquire the necessary unloading point information for transport vehicle scheduling. Subsequently, the improved ant colony algorithm is once again utilized to optimize the transport vehicle scheduling. In cases of agricultural machinery failure or changes in the operating environment, two distinct methods are employed based on the situation. The first involves double-layer rescheduling of both harvester and transport vehicles, while the second employs single-layer rescheduling exclusively for the transport vehicles, yielding the respective rescheduling results. The outcomes demonstrate that the proposed solution method effectively identifies the current optimal scheduling plan for both the harvester and transport vehicle in the event of malfunctions. Moreover, under the premise that the unproductive waiting time of the harvester is reduced to zero, and the number of transport vehicles is minimized, it achieves the minimization of operating time cost and transportation cost. This method exhibits significant potential for seamless integration into the practical application of unmanned farms, providing a foundation for addressing scheduling and management challenges in multi-agricultural machinery collaborative operations within complex farmland operating environments.

Smart agriculture has taken center stage in the contemporary landscape of agricultural science and technological competition, marking a pivotal trajectory for the advancement of modern agriculture. The dynamics of agricultural management in our nation have undergone a profound transformation with the advent of smart agriculture, transitioning from traditional methods employed by small-scale farmers with modest plots to embracing larger-scale frameworks, including professional cooperatives, family farms, and leading enterprises. This transformative journey has heightened the precision and systemization of farmland operations, making multi-machine collaborative operations an indispensable part of it. However, the complexity and dynamics inherent in large-scale farmland operations introduce formidable challenges[1]. The failure of a single agricultural machine or alterations in the operational environment can disrupt the entire planning task. Consequently, orchestrating the seamless collaboration of multiple agricultural machines and promptly and efficiently adapting the operational plans of each machine in response to unforeseen changes stand as paramount challenges in the realm of multi-machine collaborative operations within complex farmland environments.

The realization of smart agriculture is intricately linked to the crucial support of advanced agricultural machinery and equipment. Despite China's enhanced commitment to full-spectrum mechanization in recent years, the nation's agricultural mechanization and the intelligence level of its farming equipment still lag behind that of developed countries. Farmland operations primarily rely on the deployment of the same type of agricultural machinery. While this method is relatively mature, the scene setting is simplistic. Maoudj et al.[2] proposed a task allocation and scheduling method based on the contract network protocol and priority rules, which can minimize the total execution time of the task while satisfying all task constraints. Conesa-Munoz et al.[3] calculated vehicle cover crop trajectories with different characteristics based on the simulated annealing

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algorithm. In the same year, they proposed integrating the Mix-opt operator into the algorithm to optimize the operation paths of multiple agricultural machinery[4]. Seyyedhasani et al.[5] applied the enhanced Clark-Wright algorithm and the tabu search algorithm to optimize the operational paths of multiple agricultural machinery, resulting in a notable 32% reduction in operation completion time. Yao et al.[6] proposed an innovative conflict-free cooperative operation path optimization algorithm for multiple combine harvesters, effectively reducing operation time in both rectangular and trapezoidal farmland. Simultaneously, researchers have delved into collaborative operation allocation of agricultural machinery with different functions. Bochtis and Sorensen[7,8] introduced the concepts of service units (SUs) and primary units (PUs) for addressing the vehicle routing problem with time windows in the context of collaborating heterogeneous agricultural machines. When engaged in field operations, SUs or a team of SUs are essential to meet requests for on-site services from PUs. A theoretical approach was employed to tackle the planning and scheduling tasks of the SUs, determining optimal paths for their operations. Jensen et al.[9] proposed a path planning method within and between fields for one-grain truck to unload grain from multiple harvesters, to minimize the operation cost of agricultural machinery. Zhang et al.[10] studied the master-slave navigation system of two tractors used in agricultural field work. They reported that the two tractors could work together safely, and the efficiency of the master-slave navigation system was 95.1% higher than that of traditional single-machine work. Li et al.[11] investigated a follow-up master-slave navigation system for agricultural machineries, in which the slave machine tracked the host operation,and realized the master-slave navigation cooperation initially. With the goal of the shortest operation time, best operationefficiency, and economy, Li et al.[12] proposed an intelligent scheduling method for multi-machine cooperative operation based on NSGA-III and improved ant colony algorithm.

Dynamic farm job allocation stands out as a formidable challenge within the realm of multi-machine collaborative task distribution. It refers to the process of adjusting the job plan for a fleet of farm machinery in response to unexpected events that prevent the continuation of the original operation plan[13]. Adjustments are made based on the progress of the ongoing tasks and the locations of the machinery, ensuring that the equipment can proceed with and complete a new set of tasks. Despite its significance, research on dynamic task allocation remains relatively scarce. Per-Olof Gutman[14] employed the optimal control feedback algorithm to dynamically replan routes online when autonomous tractors faced delays or malfunctions during field operations. Seyyedhasani et al.[15] introduced an application method for the vehicle routing problem, enabling dynamic route recalculations in response to the unpredictable scenarios involving multiple machines within a single field. Cao et al.[16] delved into multi-machine collaborative operation task planning using the ant colony algorithm, aiming to realize efficient scheduling and management of collaborative navigation operations amidst the dynamic farmland environment. Wang and others[17] introduced a method for dynamic task allocation for homogeneous farm machinery fleets based on an improved contract net algorithm, addressing issues such as increased operation time and incomplete tasks due to unforeseen circumstances. That same year, taking the total operation time as the comprehensive optimization objective, Liang et al[18]. proposed an improved iterative greedy method for multi-machine dynamic path planning for wheeled autopilot tractors, to solve the problem of low efficiency or even failure of path planning results of traditional methods after uncertainty.

While domestic and international scholars have made certain strides in the two domains of multi-machine collaborative operation and dynamic job allocation, there remain three unresolved issues. Firstly, in response to the increasing intricacies of agricultural research, path planning within the field must now factor in complex scenarios. Secondly, the existing body of research on collaborative scenarios predominantly centers on multi-machine collaboration and job allocation methods, overlooking the integration of rescheduling after dynamic situations occur into the model. Finally, when employing various algorithms to address job allocation problems, prior research has primarily concentrated on enhancing the algorithm's performance, such as improving efficiency and resolving local optimal problems. This focus often neglects the considerations of actual scenarios in improving the algorithm for practical applications.

In this study, the co-optimization objectives aim to minimize the completion time of the latest harvester, reduce the total distance covered by the transporter while ensuring the harvester's unproductive waiting time is zero, and minimize the number of transporters. Simultaneously, the study seeks to determine the optimal scheduling plan for both the harvester and transporter under various dynamic scenarios. A dynamic allocation method, based on an improved ant colony algorithm for the cooperative operation of harvesters and transporters, is proposed in two steps. Firstly, the study outlines the scenarios for multi-machine cooperative job allocation. Secondly, it introduces a job allocation model at both static and dynamic levels: initially, the improved ACO algorithm is utilized to derive the optimal plan for harvester scheduling. This plan is then fed into the data transformation algorithm to obtain unloading point information necessary for transporter scheduling. Finally, the improved ACO algorithm is employed once more to solve the optimal plan for transporter scheduling. In the event of agricultural machinery failure or changes in the operating environment, two approaches are employed based on the circumstances. One involves double-layer rescheduling for both harvesters and transporters, while the other entails single-layer rescheduling exclusively for transporters, resulting in the rescheduling outcomes. In addition, the article also improves the ant colony algorithm at the algorithmic level for the problems that exist when solving the model to be more adapted to the scenarios of multi-machine collaborative dynamic job assignment.

Methods

Scenario description of multiple agricultural machinery cooperation

Collaborative multi-machine operations play a pivotal role in enhancing the utilization efficiency of agricultural machinery and optimizing the productivity of large-scale agricultural machinery groups[1]. Several operating modes exist for multi-machine collaborative operations in the field, including those that necessitate master-slave cooperation and those that operate independently without collaboration[19]. In cooperative modes, maintaining



Fig. 1. Distribution map of wheat fields.

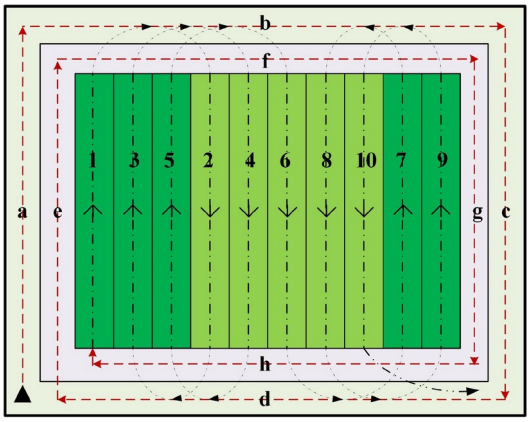


Fig. 2. Harvester field path planning map.

a relatively constant distance and declination angle between the slave and host machines is typical. This study employs a wheat harvester as the primary machine and a grain transport truck as the secondary machine, utilizing the second operating mode. When the warehouse of the wheat harvester reaches full capacity, the transport vehicle must proceed to the harvester location for coordinated grain loading. Although ongoing collaboration is unnecessary, the transport vehicle still requires computation of detailed status information for each grain unloading point, based on the harvester's scheduling results, for effective dispatch. Throughout the collaboration process, it is imperative to integrate path planning constraints and time window constraints. This integration facilitates the formulation of precise and detailed scheduling plans for both harvesters and transport vehicles, optimizing their scheduling objectives and consequently enhancing the overall efficiency of the model. The study takes the wheat harvesting and transportation scenario at Huaxing Farm as an example. The 15 wheat fields involved are illustrated in Fig. 1.

Problem definition

The farm consists of farmland of various sizes, with a central warehouse. The coordinates for both the warehouse and each farmland are known. Multiple harvesters of the same type and transport trucks are available. The road speeds of both harvesters and transport trucks, as well as the operating speed of the harvesters, are known. Initially, all harvesters and transport vehicles are stationed at the warehouse. All harvesters and transport vehicles on the farm are currently being deployed to the farmland for harvesting and transportation operations. The harvesters and transport vehicles are required to follow the planned route within the field and use point-to-point routes outside the field. The harvesting route is optimized with the goal of minimizing the completion time of the latest finished harvester. Transportation is optimized to minimize the total transportation distance of the vehicles, considering zero unproductive waiting time for the harvester and minimizing the number of

transport vehicles. Specifically, in the event of agricultural machinery breakdowns or changes in the operating environment, rescheduling of harvesters and transport vehicles is necessary to determine the current optimal solution.

Related symbols and variable descriptions

- Farmland variables comprise the warehouse F0, with coordinates F0_C = (x0, y0), and the farmland set F, where F = 1, ..., N, with N being the number of farmlands. The farmland coordinate set F_C, $F_C_i = ((x_{i1}, y_{i1}), (x_{i2}, y_{i2}), (x_{i3}, y_{i3}), (x_{i4}, y_{i4}))$, i = 1..., N, denotes the coordinates of the four vertices of each farmland in the following order: upper-left, upper-right, lower-right, and lower-left vertices.
- Agricultural machinery variables encompass the harvester set H, where H = 1, ..., M, with M being the number of harvesters. It includes the operating speed of the harvester v_f, the road speed of the harvester v_r, the efficiency of the harvester E, harvester granary capacity c_h, and harvester cutting width B. Additionally, there is the transport vehicle set S = 1, ..., m, with m being the number of transport vehicles, including transport vehicle speed v_s and transport vehicle granary capacity c_s.
- Path transfer variables involve the transfer time FT between farmlands and the transfer time DT from the warehouse to each farmland.
- Time variables comprise farm operation time T, initial time IT, time ET during dynamic situations, demand point service time ST (recording the harvester's full warehouse status each time), and warehouse service time DST.

Objective function

This study commences with multi-machine collaborative operations, exploring the formulation of collaborative scenarios and the handling of diverse dynamic situations within such scenarios. Subsequently, it establishes a collaborative dynamic operation allocation model between harvesters and transport vehicles. The model comprises two steps. Firstly, a two-level scheduling model for harvesters and transporters is established. The upper level involves harvester scheduling, where the optimal scheduling sequence of harvesters is determined by optimizing the completion time of the latest completed harvester (objective function f_1, as shown in formula (1). The lower level is dedicated to transport vehicle scheduling. Given the condition that the unproductive waiting time of the harvester is 0, the model seeks to minimize the number of transport vehicles while subsequently minimizing the total distance covered by the transport vehicles. The corresponding objective function is f_2, as indicated in formula (4). The second step involves establishing a double-layer and single-layer rescheduling model for harvesters and transporters. The primary objective is to optimize harvester and transporter dispatching goals in response to the current dynamic situation, ensuring the overall planning is effective and as efficient as possible. The corresponding objective function is f_3, outlined in formula (6). In formula (1), wt_i represents the total operating time of harvester H_i, and tt_i signifies the total transfer time of harvester H_i.

$$f_1 = min\left\{max\left\{\bigcup_{i=1}^{M}(wt_i + tt_i)\right\}\right\} \tag{1}$$

$$wt_i = \sum_{j=1}^{N} T_j \cdot x_{i,j} \tag{2}$$

$$tt_i = \sum_{p=1}^{N} DT_{1,p} \cdot x_{ip} + \sum_{l=1}^{N} DT_{1,l} \cdot x_{il} + \sum_{p=1}^{N}\sum_{k=1}^{N} FT_{p,k} \cdot x_{ipk} \tag{3}$$

In Eq. (2), When farmland F_j is in the job assignment sequence of harvester H_i, x_{ij} = 1, 0 otherwise. In Eq. (3), $DT_{1,p}$ is the transfer time of Harvester H_i from the warehouse to the first operational farmland, $DT_{1,l}$ is the transfer time of Harvester H_i from the last operational farmland to the warehouse, and $FT_{p,k}$ is the transfer time of Harvester H_i from the pth farmland to the kth farmland. when farmland F_p is the first farmland in the job assignment sequence of harvester H_i, x_{ip} = 1, 0 otherwise; when farmland F_l is the last farmland in the job assignment sequence of harvester H_i, x_{il} = 1, 0 otherwise; when farmland F_p and farmland F_k are the 2 farmlands in the job assignment sequence of harvester H_i executed in sequence, $FT_{p,k}H_i$ then x_{ipk} = 1, 0 otherwise.

$$f_2 = min\left\{\bigcup_{i=1}^{m} td_i\right\} \tag{4}$$

$$td_i = \sum_{p=1}^{D} dt_{1,p} \cdot y_{ip} + \sum_{l=1}^{D} dt_{1,l} \cdot y_{il} + \sum_{p=1}^{D}\sum_{k=1}^{D} ft_{p,k} \cdot y_{ipk} \tag{5}$$

In Eq. (5), td_i denotes the total transportation distance of the transporter S_i and D is the number of demand points. As in Eq. (3), $dt_{1,p}$, $dt_{1,l}$, and $ft_{p,k}$ denote the time of the transportation vehicle S_i from the warehouse to the next demand point, the time from the demand point to the warehouse before returning to the warehouse, and the time from the pth demand point to the lth demand point, respectively. And the definitions of y_{ip}, y_{il}, and y_{ipk} are analogous to the definitions of x_{ip}, x_{il}, and x_{ipk} in Eq. (3).

The objective function of the whole model under the premise of avoiding planning failure is shown in Eq. (6).

$$f_3 = f_1 + f_2 \tag{6}$$

Establishment of static collaborative job allocation model

The collaboration between harvesters and transport vehicles is a common occurrence in large-scale agricultural endeavors, particularly in the context of crop harvesting. Crops such as wheat, cotton, and corn necessitate the coordinated efforts of harvesters, transport vehicles, and other machinery. Nevertheless, the progress of multi-machine collaborative technology has been relatively slow in recent times. There is an urgent need to establish a comprehensive, systematic, and adaptable multi-machine collaborative operation allocation model capable of addressing complex and dynamic scenarios. The static collaborative work allocation method for harvesters and transport vehicles, proposed in this study, is outlined in three distinct steps. The framework is visually represented in Fig. 3.

Harvester scheduling

Due to the diverse shapes and sizes of individual farmlands and the varying paths taken by the harvester, the operational time for each farmland differs. Furthermore, efficient scheduling of subsequent transport vehicles necessitates detailed path segments and coordinates for demand points when the harvester warehouse reaches full capacity at a specific time. To facilitate subsequent calculations, the irregular farmlands are abstracted into regular rectangular shapes. And according to literature[20,21], the path planning for these farmlands is conducted with the longest side serving as the working direction. The length L_i and width W_i of farmland F_i are determined using Formulas (7) and (8).

$$L_i = max \left\{ \begin{array}{l} max\left\{d\left(F_C_{i1}, F_C_{i2}\right), d\left(F_C_{i3}, F_C_{i4}\right)\right\} \\ max\left\{d\left(F_C_{i2}, F_C_{i3}\right), d\left(F_C_{i4}, F_C_{i1}\right)\right\} \end{array} \right\} \tag{7}$$

$$W_i = min \left\{ \begin{array}{l} max\left\{d\left(F_C_{i1}, F_C_{i2}\right), d\left(F_C_{i3}, F_C_{i4}\right)\right\} \\ max\left\{d\left(F_C_{i2}, F_C_{i3}\right), d\left(F_C_{i4}, F_C_{i1}\right)\right\} \end{array} \right\} \tag{8}$$

Take planting crops in the turning area of the field as an example. To achieve complete coverage of the working area, a combination of straight travel and detour is employed for path planning[17]. Upon entering the farmland entrance, the harvester initiates two circles around the edge: $a->b->c->d->e->f->g->h$[12,22]. Subsequently, cross-row operations are performed, with a loop every 10 rows. In each cycle, the harvester goes up for odd-numbered sections and down for even-numbered sections. Based on this approach, all path segments within the farmland are systematically numbered. The harvester's path planning in the field is illustrated in Fig. 2.

During operation, when the remaining width of the field is less than the operating width of the agricultural machinery, specific actions are taken. In the case of a seeding machine, to prevent redundant operations, smaller machinery is utilized for cultivation, or the number of seed rows in adjacent working rows is adjusted accordingly.

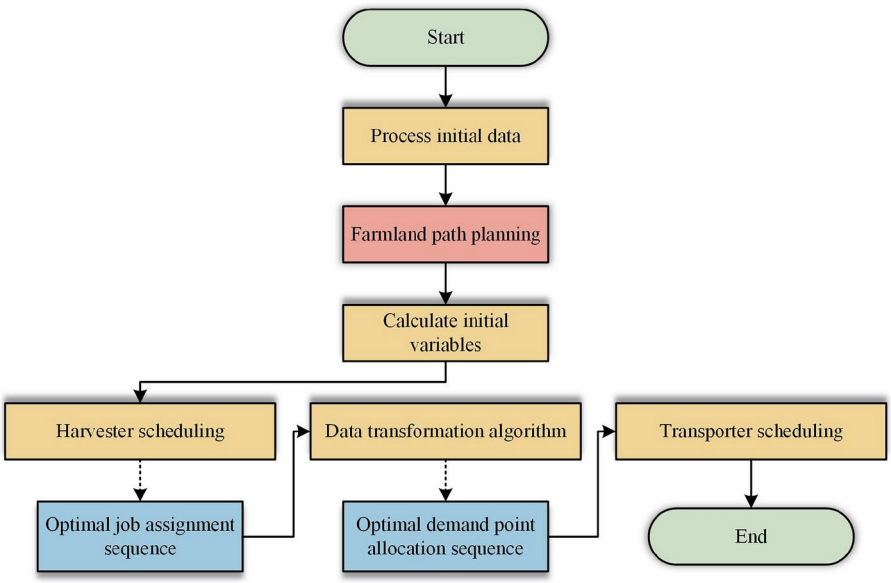


Fig. 3. Static collaborative work allocation framework.

However, this study focuses on harvesting machinery, and as such, we treat fields that are not wide enough as a single row. The total number of longitudinal paths Z for the harvester in the field is given by formula (9).

$$Z = \left\lfloor \frac{W}{B} \right\rfloor \tag{9}$$

The path segments from boundary a to boundary h, involved in the two detours, need to be calculated separately. Therefore, the number of remaining longitudinal paths is Z-4, and the length is $L_i - 4B$. If the turning point is approximated as a right angle and its additional time consumption is not considered, Formula (10) calculates the harvester's operating time for each farmland. In this formula, T_i denotes the operating time of the harvester in farmland F_i, where i ranges from 1 to N, and N represents the total number of farmlands.

$$T_i = \frac{(a+b+c+d+e+f+g+h) + (Z-4) \cdot (L_i - 4B)}{v_f} \tag{10}$$

Each field is operated by a single harvester, as follows:

$$\sum_{i=1}^{M} x_{ij} = 1 \tag{11}$$

All harvesters are mandated to complete all fields, as detailed below:

$$\sum_{i=1}^{M} \sum_{j=1}^{N} x_{ij} = N \tag{12}$$

Given that the aforementioned constraints are satisfied, the optimal sequence for harvester operation allocation that minimizes the objective function f_1 is determined by considering the calculated transfer time between farmlands, the transit time from the warehouse to each farmland, and the operational time on each farmland. This optimization problem resembles a variant of the Traveling Salesman Problem (TSP)[23]. In this analogy, the coordinates of each farmland correspond to the cities' locations, and the harvester is likened to the traveling salesman. The traveling salesman embarks from a starting point, visits all the cities, and ultimately returns to the origin. The objective is to minimize the total travel cost. Owing to the fact that there are multiple harvesters involved, the problem can be regarded as a Multiple Traveling Salesman Problem (MTSP). The MTSP is known to be an NP-Hard problem[24], which is typically addressed using modern heuristic algorithms. In this study, the ant colony optimization algorithm is employed to determine the most efficient job allocation sequence for the harvesters.

Data conversion algorithm

In the following section, the scheduling of transport vehicles requires detailed information about the numbers, times, specific path segments, coordinates, and other relevant details of all demand points. Consequently, the data conversion algorithm must be employed to transform the optimal job allocation sequence of the harvester, as obtained in the previous section. The calculation of farmland F_i yield is presented in formula (13), where e is the efficiency (unit: mu/h), and acre_yield is the wheat yield per acre.

$$O_i = T_i \cdot e \cdot acre_yield \tag{13}$$

The time for harvester H_i to reach each farmland in its job allocation sequence is illustrated in Formula (14).

$$\left\{ \begin{array}{l} AT_{i,j} = IT + DT_j, \quad j \quad is \quad the \quad first \quad working \quad farmland \quad of \quad H_i. \\ AT_{i,j} = AT_{i,j-1} + T_{j-1} + FT_{j-1,j}, \quad j \quad is \quad H_i \quad other \quad working \quad farmland. \end{array} \right\} \tag{14}$$

Design an algorithm for data conversion to compute the farmland number and demand time at the location of each demand point, grounded in the yield of individual farmlands and the arrival time of the harvester at each farmland. Specifically, if the last bin of each harvester contains less grain than the granary's capacity, no demand point is established. Instead, the harvester transports the surplus back to the warehouse for unloading.

By comparing the demand time of each demand point with the start and end times of each path segment within the farmland containing the demand point, we can determine the specific path segment where the demand point is situated. Following the delineation of specific path segments in field path planning, the disparity between the identified segment and the coordinates of the farmland entrance is calculated. Subsequently, the difference between the demand point and the starting or ending points of the path segment is computed, enabling the precise determination of the coordinates of the demand point.

Transporter scheduling

Each farmland contains one or more demand points, allowing multiple transport vehicles to visit a single farmland. However, each demand point can only be visited by one transport vehicle, as follows:

$$\sum_{j=1}^{N} \varepsilon_{i,j} \geq 1 \tag{15}$$

$$\sum_{i=1}^{m} \sum_{k=1}^{D} \omega_{i,k} = 1 \tag{16}$$

Among them, $\varepsilon_{i,j}$ indicates the presence of farmland F_j in the task sequence of transport vehicle S_i, and $\omega_{i,k}$ indicates the inclusion of demand point D_k in the demand point sequence of transport vehicle S_i. Time requirements are essential for transporter scheduling. To ensure that the harvester incurs no unproductive waiting time, the transporter must reach the harvester at the specified demand point, as follows:

$$S_T_{i,j} \leq D_T_j \tag{17}$$

$S_T_{i,j}$ represents the time at which transport vehicle S_i arrives at demand point D_j, while D_T_j denotes the demand time at demand point D_j. The transport trucks initiate their journey from the warehouse, adhering to capacity constraints. Following the transportation of a specific number of demand points, each transport vehicle is required to return to the warehouse.

Due to time window constraints and capacity limitations, transporter scheduling in this study is a variant of the Vehicle Routing Problem (VRP) known as CVRPTW (Vehicle Routing Problem with Time Window and Capacity Constraints)[25]. In contrast to the traditional CVRPTW, the time for each demand point in this paper is fixed, requiring transport vehicles to reach the demand point before the designated time. Additionally, the Ant Colony Algorithm is employed to optimize the task allocation sequence for transport vehicles. Unlike harvester scheduling, where the number of harvesters remains constant, transporters must select the minimum number required to fulfill the current transportation task. If the algorithm can identify a feasible solution with m transport vehicles, m is determined as the minimum number needed.

Establishment of dynamic collaborative job allocation model

In the process of allocating resources for multi-machine collaborative operations, various uncertain factors, such as agricultural machinery failure or the emergence of new operational tasks, may arise due to changes in machinery and the natural environment. These uncertainties play a pivotal role in determining the successful completion of multi-machine collaborative operations. Addressing the challenge of reallocating agricultural machinery to optimize its objectives when uncertain factors arise has long been a persistent issue for agricultural machinery cooperatives and farms. Drawing from the collaborative operation scenario analysis of harvesters and transport vehicles presented in literature[26], the path planning scheme can be dynamically adjusted in a timely manner to address six distinct dynamic situations.

(1) Harvester failure
(2) Increase in harvesters
(3) Decrease in farmland
(4) Increase in farmland
(5) Transport vehicle failure
(6) Increase in transport vehicles

The six aforementioned dynamic situations are categorized based on the perspective of rescheduling implementation. The first four types involve a double-layer rescheduling of harvester transport vehicles, while the latter two involve a single-layer rescheduling of transport vehicles. The dynamic collaborative work reallocation framework of harvesters and transport vehicles is illustrated in Fig. 4.

Double-layer rescheduling of harvester transport vehicles

The harvester H_h breaks down at ET time. In the case of temporary failures such as an urgent need for refueling or film replacement, the affected field remains untreated until the faulty machine is repaired. Only the farmland where other agricultural machinery is situated and the farmland not in operation will be rescheduled. In the event of a permanent fault, such as damage to the harvester, it is assumed that external force is employed to remove the faulty agricultural machinery, necessitating action on the affected farmland. This article focuses on the study of permanent faults, and the rescheduling steps are outlined as follows:

The initial step involves calculating the farmland on which each harvester is currently operating or the farmland it is poised to work on at the time of failure. Subsequently, the identified farmland is labeled as new farmland. The formulation of Formula (18) signifies that harvester H_i is in the transition stage from farmland F_j to farmland F_{j+1}, with F_{j+1} designated as the upcoming operational farmland. Formula (19) establishes that harvester H_i is working in farmland F_j.

$$AT_{i,j} + T_j \leq ET \leq AT_{i,j+1} \tag{18}$$

$$AT_{i,j} \leq ET \leq AT_{i,j+1} - FT_{j,j+1} \tag{19}$$

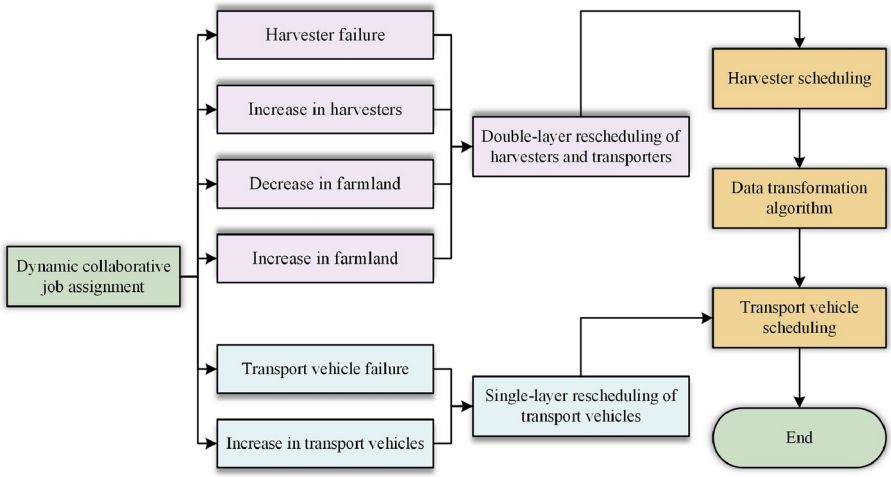


Fig. 4. Dynamic collaborative work reallocation framework.

The second step involves updating the operation time of the newly identified farmland. Alongside the new farmland, unoperated farmland is consolidated into the new farmland set, which serves as input for subsequent rescheduling. Formulas (20) represent the update formulas for the operation time of the new farmland selected in the previous transfer stage and operation stage.

$$\begin{cases} T_{new} = T_{new} + (AT_{i,new} - ET) \\ T_{new} = T_{new} - (ET - AT_{i,new}) \end{cases} \tag{20}$$

The third step involves a detailed calculation of the specific location of the faulty harvester. Given the adoption of a cross-row harvesting method, ensuring the crops remain undamaged necessitates that other harvesters cannot employ a point-to-point strategy from the entrance of the faulty field to the fault point. Different methods are employed to calculate the time from the fault point to the entrance of the faulty farmland for various boundary and path segments where the fault point is located. This calculated time is then utilized to update the transfer time from the faulty farmland to each farmland. Consequently, the faulty farmland is treated as a new farmland during subsequent rescheduling. In Formula (21), $FT_{i,j}$ represents the transfer time from the faulty field to other farmland, where j = 1,...,N, and the TIME calculation is contingent on the actual road section.

$$FT_{i,j} = FT_{i,j} + time, i = fault_field, \forall j \in N \tag{21}$$

The fourth step involves utilizing the new farmland set, the updated operation time for each farmland, the revised transfer time between farmlands, and the remaining harvesters for rescheduling. This is accomplished using the ant colony algorithm from the time of failure to acquire a new job allocation sequence. Notably, to minimize harvester transfer time, the first farmland of all harvesters is set as the new farmland selected in the initial step.

The fifth step involves determining the granary capacity of each harvester at the time of failure, serving as the initial cache capacity for each harvester. The new job allocation sequence and updated time variables are inputted into the data conversion algorithm to acquire new demand point information. Formulas (22) are the calculation formulas for the granary capacity C_{i,c_h} when the last demand point of the harvester H_i is in the current farmland, the previous farmland, and the fault point is in the transfer stage at the time of failure. Among them, $D_{i,j-1}$ is the demand time of the previous demand point D_{j-1} of the harvester H_i, k and l are the previous farmland and the current farmland respectively.

$$\begin{cases} C_{i,c_h} = c_h - (ET - D_{i,j-1}) \cdot e \cdot acre_yield \\ C_{i,c_h} = c_h - (ET - D_{i,j-1} - FT_{k,l}) \cdot e \cdot acre_yield \\ C_{i,c_h} = c_h - (AT_{i,k} - T_k - D_{i,j-1}) \cdot e \cdot acre_yield \end{cases} \tag{22}$$

In the sixth step, the remaining granary capacity of each transport vehicle at the time of failure is employed as the initial granary capacity for each transport vehicle. Due to the presence of a demand moment at the demand point, synchronization is maintained between the harvester and transporter. Demand points occurring before the fault time have already been transported, while those after the fault time have yet to commence transportation. Consequently, the position of each transport vehicle at the time of failure is utilized as the initial position, the time of failure is adopted as the initial time, and only the new demand points are transported.

Similarly, increases in harvesters can take two forms. One type involves agricultural machinery with temporary malfunctions, which can resume operation directly after repair. The other type involves newly introduced agricultural machinery from the warehouse, requiring operation starting from the warehouse. The study takes the second form as an example, which is the same as the above-mentioned harvester failure processing. Since it doesn't involve dealing with faulty fields, the third step can be omitted. The distinction lies in the calculation of the completion time for each harvester. For the newly added machine, the transfer time from the warehouse to the first field is included, while this is not applied to other harvesters.

In situations where new farmland is added or original farmland is reduced at a specific time, the processing method is relatively straightforward. It involves modifying the unvisited farmland list. The third step is unnecessary in this scenario.

Single-layer rescheduling of transport vehicles

If the relevant parameters of the harvester and farmland remain unchanged, the harvester will continue its operations based on the original optimal job allocation sequence. Additionally, the relevant information regarding demand points, obtained through the data conversion algorithm, remains constant. Consequently, only the transport vehicle portion in the collaborative model requires separate rescheduling. Two dynamic situations may arise for transport vehicles: transport vehicle failure and the addition of transport vehicles. The subsequent analysis focuses on the example of transport vehicle failure, with the method applicable to the increase in transport vehicles.

The first step involves analyzing the status of each transport vehicle at the time of failure, categorized into four states: (1) On the way to the demand point; (2) On the way to the warehouse; (3) Loading grain at the demand point; (4) Unloading grain at the warehouse. This entails calculating the coordinates of each transport vehicle at this moment and the number of the next demand point. For the first two states, the transfer time from the current location to the next demand point is updated, while the latter two states do not require processing. The calculation of the coordinates of the transport vehicle failure moment is shown in formula (23). Among them, $D_{i,j-1}$ is the demand time of the last demand point of transport vehicle S_i. (x, y), (x_1, y_1), and (x_2, y_2) are the coordinates of the transport vehicle at the time of failure, the coordinates of the previous demand point j-1, and the coordinates of the next demand point j.

$$\begin{cases} x = x_1 + (ET - D_{i,j-1}) \cdot v_s \cdot \cos \arctan \frac{y_2 - y_1}{x_2 - x_1} \\ y = y_1 + (ET - D_{i,j-1}) \cdot v_s \cdot \sin \arctan \frac{y_2 - y_1}{x_2 - x_1} \end{cases} \tag{23}$$

In the second step, the choice of the initial demand point for each transport vehicle is predetermined as the original next demand point. Subsequently, the ant colony algorithm is applied to reassign the demand points following the approach outlined in the previous section.

The third step involves verifying the possibility of identifying a new, feasible demand point allocation sequence. If such a sequence cannot be found, it suggests that the current number of transport vehicles is insufficient, necessitating the deployment of additional vehicles.

Algorithm

The static model for multi-machine collaborative dynamic job allocation comprises three modules: harvester scheduling, data conversion algorithm, and transporter scheduling. Both harvester and transporter scheduling are based on the ant colony algorithm, while the data conversion algorithm translates the harvester job allocation sequence into the demand point information needed for transporter scheduling. The dynamic model includes two modules: double-layer rescheduling for both harvesters and transporters, and single-layer rescheduling for transporters. Double-layer rescheduling employs the ant colony algorithm and data conversion algorithm, whereas single-layer rescheduling relies solely on the ant colony algorithm. The algorithm flow chart is depicted in Fig. 5. We provide a detailed introduction to the ant colony algorithm and its modifications, along with an explanation of the data conversion algorithm's implementation.

Basic ant colony algorithm

The ant colony algorithm is a probabilistic algorithm employed for discovering optimal paths. It boasts features such as distributed computing, positive information feedback, and heuristic search. With robust global optimization capabilities and ease of implementation, this algorithm finds widespread application in diverse domains, including path planning, combinatorial optimization, and scheduling[16,27,28]. Nevertheless, the ant colony algorithm does exhibit certain drawbacks, and two of these become particularly conspicuous in the cooperative dynamic job allocation model involving harvesters and transporters.

(1) The study concentrates on large-scale farms, characterized by a significant fleet of harvesters and transport vehicles. In employing the basic ant colony algorithm for job allocation, each ant symbolizes a solution for path selection, but this approach falls short in capturing the diversity inherent in managing multiple agricultural machines.

(2) In the process of solving the optimal demand point allocation sequence of transport vehicles, since transport vehicles have capacity constraints and demand points have time constraints. When selecting demand points, the ant colony selects the demand point with a larger current probability (related to the pheromone concentration between demand points and the transfer time between demand points) based on roulette. When selecting, the settings of the pheromone factor and the heuristic factor are not well balanced, which often makes the demand time of the selected demand point too late, causing the transport vehicle to reach the termination

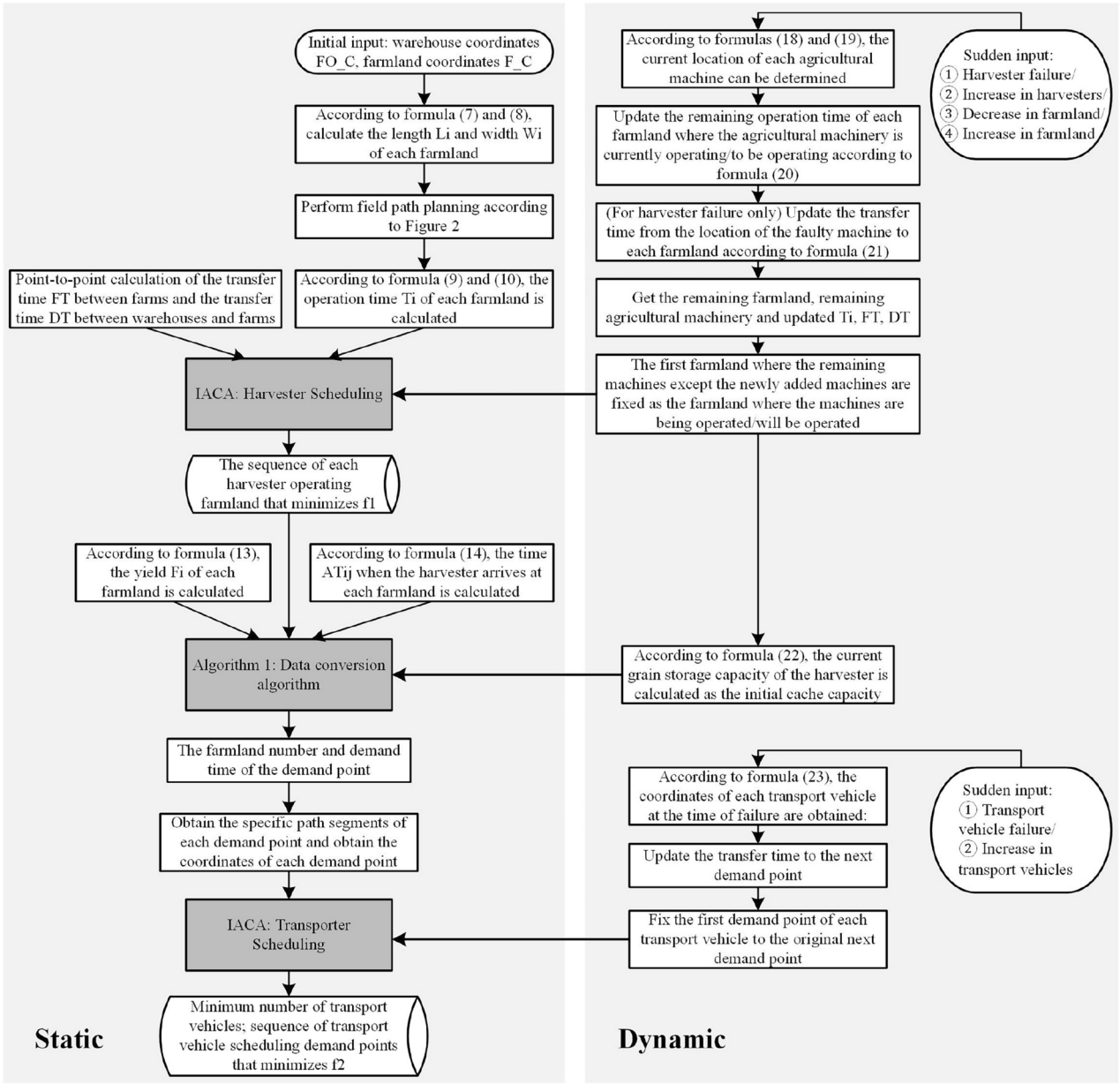


Fig. 5. Flowchart of multi-machine collaborative dynamic job allocation algorithm.

time after selecting a small number of demand points. Each time there will be several earlier demand points that are not selected and cannot be selected because they are too early, making it difficult to find a feasible solution that includes all demand points. After multiple iterations, the continuous accumulation of pheromones causes the algorithm to converge on an infeasible solution, resulting in the failure of the transport vehicle scheduling task.

Improved ant colony algorithm

To address the first issue, a solution for multi-level cooperation in ant colonies is proposed: the introduction of multi-level division and path selection. This involves splitting each ant into several smaller ants. Each small ant independently chooses its own path, and their paths are subsequently merged into one comprehensive path for the ant. Additionally, parallel programming technology is introduced, enabling these smaller ants to execute the path selection function concurrently in multiple threads. This improvement offers three key advantages:

(1) Different ants, representing various agricultural machines, can explore distinct paths independently. This facilitates a more comprehensive search for optimal solutions.

(2) Multiple agricultural machines can concurrently select paths, expediting the algorithm's convergence speed. The competition between different paths and the diffusion of pheromones guide the algorithm toward a more optimal solution more rapidly.

(3) In the event that some ants choose an inappropriate path, other ants retain the opportunity to select a better path. This adaptability is crucial in overcoming the challenge of the algorithm converging to a local optimal solution.

To address problem two, we propose a Rule-Ant Colony Fusion Algorithm based on the idea of Literature[29]. In the early stages, the ant colony algorithm conducts blind searches, leading to continuous round-trip oscillations and ample room for improvement in the later search process. A rule-based approach is introduced to enhance this process: the initial selection of each transport vehicle's first demand point is fixed to the demand point with the smallest current demand time. Subsequently, the remaining demand points with smaller current times are selected as much as possible following predefined rules. The fine-tuning rule allows the calculation of better feasible solutions, utilizing these solutions as the initial path to update the initial pheromone of the ant colony algorithm. The algorithm then conducts a path search based on the ant colony approach. A comparison between the feasible solution obtained from the rules and the solution from the ant colony yields the optimal solution for the fusion algorithm. This improvement streamlines the process for transporters to discover optimal solutions, particularly when dealing with a large number of demand points.

Taking transport vehicle scheduling as an example, the demand point allocation process based on the improved ant colony algorithm is shown in Fig. 6. The specific implementation steps are as follows.

Step 1: Initialization. Initialize the demand point information, the number and capacity of transport vehicles, the cost matrix, time variables, and other input information. Also, set the relevant parameters for the ant colony algorithm: the number of ants m, the number of iterations n, the pheromone evaporation rate ρ, the pheromone weight α, the heuristic weight β, and the pheromone increment Q. The values of each parameter refer to previous papers and are adjusted and selected in combination with our specific problems[16,27,28].

Step 2: Initial Pheromone Update. Utilize a rule-based algorithm to generate p allocation sequences, subsequently updating the initial pheromone matrix of the ant colony algorithm. The rule is as follows: for each transport vehicle, select demand points in turn. Given the strict time constraints, the distance factor is disregarded, and selections are made from demand points at the front of the time window. If the demand points selected by all transport vehicles satisfy the demand time constraints, this constitutes a feasible solution. Use this solution to update the pheromone matrix. Repeat this process until p iterations have been completed, then finalize the initial pheromone matrix for the improved ant colony algorithm.

Step 3: Encoding. Use 0 to represent the warehouse and numbers 1, 2, ..., d to represent d demand points. In each iteration, there are m ants, each representing a solution. Each ant is divided into s sub-ants, with each sub-ant representing one of the s transport vehicles. Initially, create a list for each ant containing s sub-lists. Each sub-list is initialized with an element 0, indicating that the sub-ant is currently located in the warehouse.

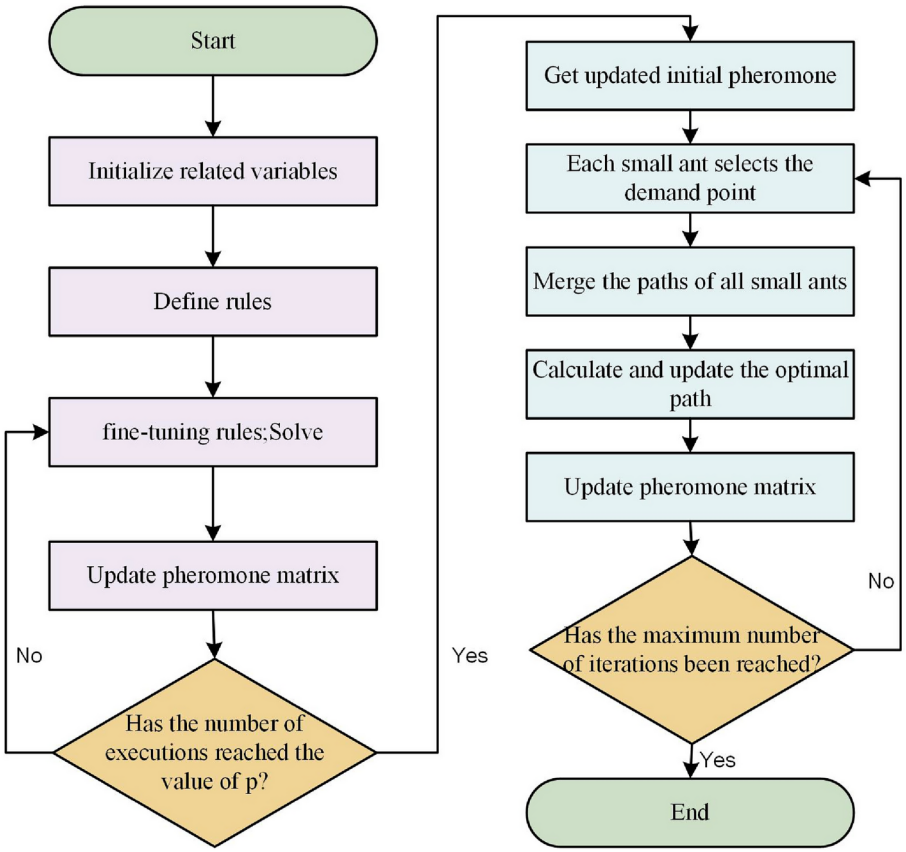


Fig. 6. Improved ant colony algorithm to solve transport vehicle scheduling process.

Step 4: Path Selection. In each iteration, select the next demand point for each sub-ant of every ant. Begin by obtaining the list of unvisited demand points J, the path list, and the remaining capacity of the sub-ant. Use formula (24) to calculate the probability of selecting each unvisited demand point, and choose the next demand point using the roulette wheel method. Add the selected demand point to the sub-ant's path list, remove it from the list of unvisited demand points, and update the remaining capacity by decrementing it by 1. If the remaining capacity of the sub-ant reaches 0, append 0 to the path list to indicate the return to the warehouse and reset the remaining capacity to the maximum value (i.e., the number of demand points a transport vehicle can carry in one trip). Here, $P_{ij}(t)$ represents the probability that the current sub-ant at demand point i selects demand point j in the t-th iteration. $\tau_{ij}(t)$ denotes the pheromone level on the edge (i,j). Dis_{ij} and Tim_j represent the distance from demand point i to demand point j and the demand time of demand point j, respectively. The smaller the sum of these two values, the larger the reciprocal, thereby increasing the heuristic value.

$$P_{ij}(t) = \begin{cases} \frac{[\tau_{ij}(t)]^{\alpha} \cdot [1/(Dis_{ij}+Tim_j)]^{\beta}}{\sum_{j \in J}[\tau_{ij}(t)]^{\alpha} \cdot [1/(Dis_{ij}+Tim_j)]^{\beta}}, j \in J \\ 0, otherwise. \end{cases} \tag{24}$$

Step 5: Decoding. After path selection, if the path of an ant is: [[0,6,5,4,0,7,12,0], [0,14,13,11,0,10,8,0], [0,9,3,1,0,2,0]] (assuming the transport vehicle can carry 3 demand points in one trip), it indicates that three sublists represent three sub-ants, meaning there are 3 transport vehicles. Each sublist represents the transportation path of one vehicle. For instance, the path of transport vehicle 1 shows that the vehicle starts from warehouse 0, visits demand points numbered 6, 5, and 4 in sequence to provide coordinated grain unloading services, then returns to the warehouse with a full load for unloading. After unloading, the vehicle starts again from the warehouse, visits demand points numbered 7 and 12, and then returns to the warehouse to complete the task.

Step 6: Calculate Total Path Distance. Consider each ant's path as a solution. If a solution lacks all demand points, set the distance of that solution to infinity, marking it as infeasible. For solutions encompassing all demand points, calculate the sum of the distance covered by all transport vehicles, defining it as a feasible solution.

Step 7: Update Optimal Path. If the total distance of a solution is less than the current shortest distance, update the shortest distance to the current distance. The optimal solution becomes the current solution.

Step 8: Update Pheromones. After each iteration, calculate the pheromone increment for each edge of the ant with the optimal solution, denoted as $\Delta\tau_{ij}$. Let ω be the set of paths traveled by the ant and L be the total distance of the ant's path. The smaller the total distance, the greater the pheromone concentration. Update the pheromone concentration of each edge to the old pheromone concentration multiplied by the volatility factor plus the pheromone increment. $\tau_{ij}(t+1)$ is the pheromone concentration on the edge (i, j) of the t+1 iteration.

$$\Delta\tau_{ij} = \begin{cases} \frac{Q}{L}, (i,j) \in \omega \\ 0, otherwise. \end{cases} \tag{25}$$

$$\tau_{ij}(t+1) = (1-\rho) \cdot \tau_{ij}(t) + \Delta\tau_{ij} \tag{26}$$

Step 9: Iterate to the Maximum Number of Times. Determine if the maximum number of iterations has been reached. If not, repeat steps 3 to 8 until the current number of iterations equals the maximum number of iterations. Then, output the optimal allocation sequence of demand points.

Step 10: Output. Based on the optimal allocation sequence of demand points, output the results in two forms: a demand point allocation operation table and a transport vehicle transfer path map.

Data conversion algorithm

The number of farmlands and demand times for each demand point are computed based on the yield of each farmland, as determined by formula (13), and the arrival time of the harvester at each farmland, as calculated by formula (14). The pseudo code for the data conversion algorithm is provided below.

```
for i ← 1 to num_harvesters do
    Initialize cache_time, cache_yield, demand_points_list, demand_points_time_list
    for j ← 0 to number_of_fields_per_harvester - 1 do
        if j = 0 then
            if yield of the first field < harvester_bin_capacity then
                Update cache_time to harvester arrival time at the first field + field operation time
                Update cache_yield to the yield of the field
            else if yield of the first field ≥ harvester_bin_capacity then
                Add index of the field to demand_points_list
                Update cache_time to cache_time + harvester arrival time at the first field + time to complete harvesting the harvester bin capacity
                Add cache_time to demand_points_time_list
                Update yield of the field to original yield − harvester bin capacity
                if remaining field yield ≥ harvester_bin_capacity then
                    Reset cache_yield and harvester arrival time, then continue to 5
                else if remaining field yield < harvester_bin_capacity then
                    Update cache_yield to remaining field yield
                    Reset cache_time to zero
                end if
            end if
        else if j ≠ 0 then
            if yield of the field + cache_yield from the previous field < harvester_bin_capacity then
                Update cache_time to original cache_time + field operation time
                Update cache_yield to yield of the field + original cache_yield
            else if yield of the field + cache_yield ≥ harvester_bin_capacity then
                Add index of the field to demand_points_list
                Update cache_time to cache_time + harvester arrival time at the field + time to complete harvesting the remaining harvester bin capacity
                Add cache_time to demand_points_time_list
                Update yield of the field to original yield − remaining harvester bin capacity
                if remaining field yield ≥ harvester_bin_capacity then
                    Reset cache_yield and harvester arrival time, then continue to 21
                else if remaining field yield < harvester_bin_capacity then
                    Reset cache_time to zero
                    Update cache_yield to remaining field yield
                end if
            end if
        end if
    end for
end for
Combine demand_points and demand_points_time, then output.
```

Algorithm 1. Data Conversion Algorithm

Experiment

Simulation conditions and parameter settings

According to the aforementioned approach, a simulation experiment on multi-machine collaborative dynamic job allocation was conducted using the improved ant colony algorithm within the Python 3.11 simulation environment. This study utilized actual data obtained from Huaxing Farm in Changji City, Xinjiang Uygur Autonomous Region. The farm comprises a warehouse and a total of 15 wheat planting fields. To facilitate path planning within the field, the four vertex coordinates of each farmland were fine-tuned to abstract it into a regular rectangular field. Subsequently, the length, width, and operation time of each farmland were calculated. For ease of subsequent coordinate calculations, the longitude and latitude coordinates of the WGS 84 coordinate system were converted into plane coordinates of the Web Mercator coordinate system. The processed farmland input information is presented in Table 1.

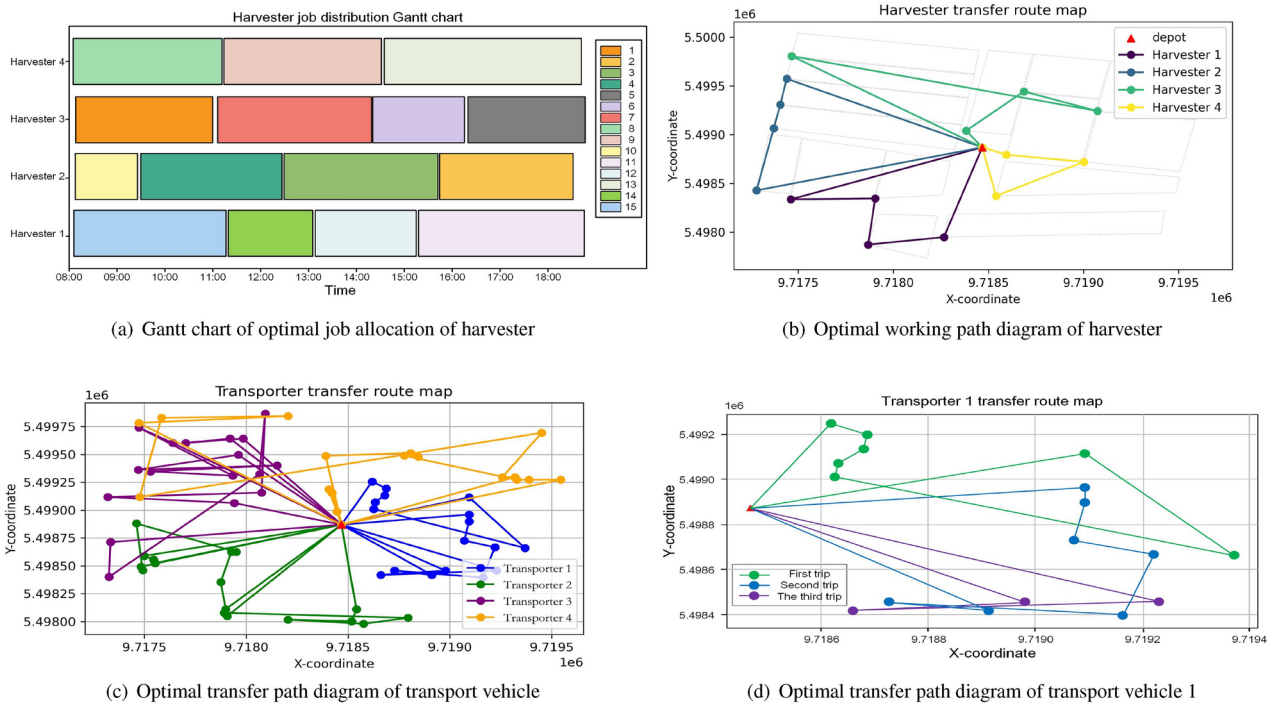
There are four John Deere S770 grain combine harvesters used for wheat harvesting at Huaxing Farm. The agricultural machinery parameters and other parameters consistent with actual operation are shown in Table 2.

Farmland number	Geographical coordinates	Length	width
1	(9717465.680833897, 5499805.419937804)	703.25	173.19
2	(9717439.406094484, 5499572.524835671)	702.57	167.6
3	(9717405.964606253, 5499308.578320465)	700.71	194.92
4	(9717371.328659888, 5499063.740795073)	701	181.99
5	(9718382.92560816, 5499039.855543105)	599.48	170.99
6	(9718686.28457891, 5499442.343882344)	290.74	278.89
7	(9719073.247826831, 5499241.69704982)	377.05	374.6
8	(9718594.919106843, 5498793.82306112)	464.04	291.13
9	(9719001.588131024, 5498719.774644955)	381.52	374.33
10	(9717280.559860291, 5498427.163771013)	416.24	131.9
11	(9717460.904114546, 5498335.798273086)	449.19	326.4
12	(9717904.59688817, 5498345.352873438)	354.11	255.48
13	(9718539.979598552, 5498371.62807587)	680.79	256.45
14	(9717868.169811198, 5497871.801803458)	313.54	243.56
15	(9718265.880955955, 5497948.238114422)	818.49	168.54

Table 1. Processed farmland input information.

Name of parameter	Value of number	Name of parameter	Value of number
Harvester operating speed	5.6 km/h	Harvester grain bin capacity	8480 kg
Harvester road speed	14.5 km/h	Transporter capacity	59360 kg
Transporter speed	40 km/h	Harvester efficiency	14000 kg/h
Warehouse latitude and longitude coordinates	(87.30248094, 44.21560927)	Initial time	8:00
Warehouse plane coordinates	(9718467.72323027, 5498869.066470771)	Warehouse service time	10 min
The width of the harvester	7.7m	Demand point service time	5min

Table 2. Parameters table.



(a) Gantt chart of optimal job allocation of harvester

(b) Optimal working path diagram of harvester

(c) Optimal transfer path diagram of transport vehicle

(d) Optimal transfer path diagram of transport vehicle 1

Fig. 7. Scheduling result graph.

Results analysis and discussion

Three types of experiments were conducted based on the provided input data. The first aimed to verify the feasibility of the static collaborative job allocation method. The second involved changing the input, adjusting the number of agricultural machinery and farmland, to validate dynamic collaborative operation allocation. The experiments demonstrated that, in collaborative scenarios with dynamic situations, jobs can be efficiently reallocated in a timely manner, allowing for the identification of optimal job allocation plans for each goal under the current circumstances. The third type of experiment focused on analyzing the optimization effect of utilizing the improved ant colony algorithm for solving multi-machine collaborative dynamic job allocation. The results affirm that, in the collaborative scenario proposed in this article, the improved ant colony algorithm proves more applicable and efficient compared to the basic ant colony algorithm.

Static collaborative job assignment

The transfer time between farmlands is derived from the harvester's road speed, the transport vehicle's speed, and the geographical coordinates of each farmland, including the warehouse. The farmland path is planned, and the operation time for each farmland is calculated based on the processed farmland input information.

The parameters of the ant colony algorithm, including the number of ants m, number of iterations n, pheromone volatilization factor ρ, pheromone weight α, heuristic weight β, and pheromone increment Q, are set to 100, 500, 0.5, 1.0, 2.0, and 800, respectively. Initially, the improved ant colony algorithm is employed for harvester scheduling, yielding the optimal job allocation sequence for the harvester: best_path=[[15, 14, 12, 11], [10, 4, 3, 2], [1, 7, 6, 5], [8, 9, 13]]. The optimal completion time is 18:48, requiring a total of 648 minutes of work. Refer to Fig. 7a,b for the corresponding job allocation Gantt chart and harvester transfer path. Utilizing best_path as the original data, the data conversion algorithm generates a total of 68 demand points across 15 pieces of farmland. Table 3 displays detailed information about these demand points (only the first 20 are listed due to space constraints). Figure 8 presents the detailed operation arrangement and demand point distribution for the arbitrary selection of harvester 2 operating in farmland 3. With the objective of minimizing unproductive waiting time for harvesters and optimizing the number of transport vehicles, the improved ant colony algorithm is once again employed for transport vehicle scheduling. In this case, the algorithm recommends the use of the minimum number of transport vehicles, i.e., 4 transport vehicles, resulting in an optimal total distance of 40.988 km. The optimal transport vehicle transport path and transport vehicle 1 transport path are visualized in Fig. 7c,d.

Dynamic collaborative job assignment

The preceding chapter's analysis highlights six dynamic situations that may arise in the collaboration between harvesters and transport vehicles. These situations are categorized into two types based on distinct processing methods: one type involves double-layer rescheduling, while the other entails single-layer rescheduling.

At the outset, both the harvester and transport vehicle adhere to the optimal job allocation sequence and demand point allocation sequence, as computed in the previous section. However, during the experimental simulation at 11:30 and 13:30, four dynamic situations unfold: harvester 1 failure, joint failure of harvesters 1 and 2, introduction of new harvester 5, and simultaneous introduction of new harvesters 5 and 6. Individual solutions

number	Required time	Farmland	Route segment	Coordinate
1	8:39	15	Boundary g	(9718541.180955956, 5498109.078114422)
2	9:15	15	Route 4	(9718516.880955955, 5498002.138114423)
3	9:51	15	Route 8	(9718204.880955955, 5498017.538114422)
4	10:28	15	Route 13	(9718576.347622622, 5497979.038114422)
5	11:04	15	Route 17	(9718795.014289288, 5498032.938114422)
6	11:42	14	Route 1	(9717891.269811198, 5498078.268470124)
7	12:18	14	Route 13	(9717898.9698112, 5498110.268470124)
8	12:54	14	Route 25	(9717906.669811198, 5498048.935136791)
9	13:32	12	Boundary h	(9717874.663554836, 5498353.052873438)
10	14:08	12	Route 11	(9717927.69688817, 5498624.752873438)
11	14:45	12	Route 22	(9717950.796888169, 5498624.752873438)
12	15:22	11	Boundary a	(9717460.904114546, 5498880.331606419)
13	15:59	11	Route 3	(9717491.704114547, 5498461.064939752)
14	16:35	11	Route 11	(9717484.004114546, 5498493.064939752)
15	17:11	11	Route 19	(9717553.304114547, 5498525.064939752)
16	17:48	11	Route 27	(9717545.604114546, 5498557.064939752)
17	18:24	11	Route 35	(9717499.404114546, 5498589.064939752)
18	8:40	10	Route 4	(9717334.459860291, 5498712.963771013)
19	9:16	10	Route 12	(9717326.75986029, 5498400.963771013)
20	9:54	4	Boundary f	(9718064.62865989, 5499322.374128407)

Table 3. Demand point information table.

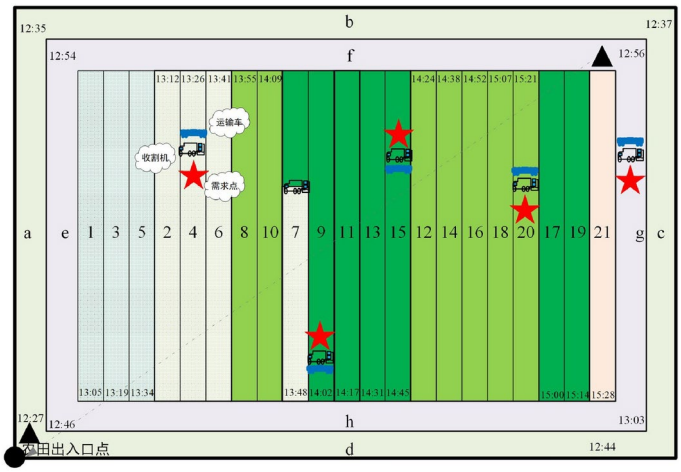


Fig. 8. Detailed operation arrangement and demand point distribution map of harvester 2 in farmland 3.

Dynamic situation	Dynamic moment	The harvester redistributes the seqence	f_1(min)	Number of demand points	Minimum number of transport vehicles	Transporter 1 Redistributes the sequence	f_2 (km)
H-1	11:30	[[3, 2, 13, 10], [6, 5, 1, 11], [8, 4, 12]]	593|21:23	46	3	[D, 0, 1, 2, 3, 4, 5, 37, D, 38, 39, 40, 41, 42, 43, 44, D, 45, 30, D]	27.154
	13:30	[[2, 1, 5], [6, 12, 11], [8, 10, 4]]	425|20:35	33	3	[D, 0, 1, 2, 3, 4, 5, 6, D, 7, 30, 31, 32, D]	19.988
H-2	11:30	[[6, 5, 4, 1, 2, 3], [8, 12, 11, 10, 13]]	872|26:02	46	2	[D, 0, 1, 2, 3, 27, 5, 6, D, 30, 31, 32, 33, 34, 35, 36, D, 37, 38, 39, 40, 41, 42, 43, D, 44, 45, D]	23.224
	13:30	[[6, 4, 5, 1, 2], [8, 12, 10, 11]]	636|24:06	33	2	[D, 0, 17, 18, 19, 20, 21, 22, D, 23, 24, 25, 26, 27, 28, 29, D, 30, 31, 32, D]	19.385
H+1	11:30	[[13, 11, 4], [3, 12], [6, 5], [8, 10], [2, 1]]	397|18:07	44	5	[D, 0, 1, 2, 3, 4, 5, 31, D, 42, 33, D]	32.221
	13:30	[[11, 1], [2, 4], [6, 5], [8, 10], [12]]	284|18:14	31	5	[D, 0, 1, 27, 28, 29, 30, 13, D]	20.458
H+2	11:30	[[13, 10], [3, 2], [6, 11], [8, 5], [4, 1], [12]]	321|16:51	44	6	[D, 0, 1, 2, 3, 4, 5, 6, D, 7, D]	23.313
	13:30	[[11, 4], [2, 5], [6, 1], [8], [12], [10]]	257|17:47	29	6	[D, 0, 1, 20, 21, 22, D]	22.404

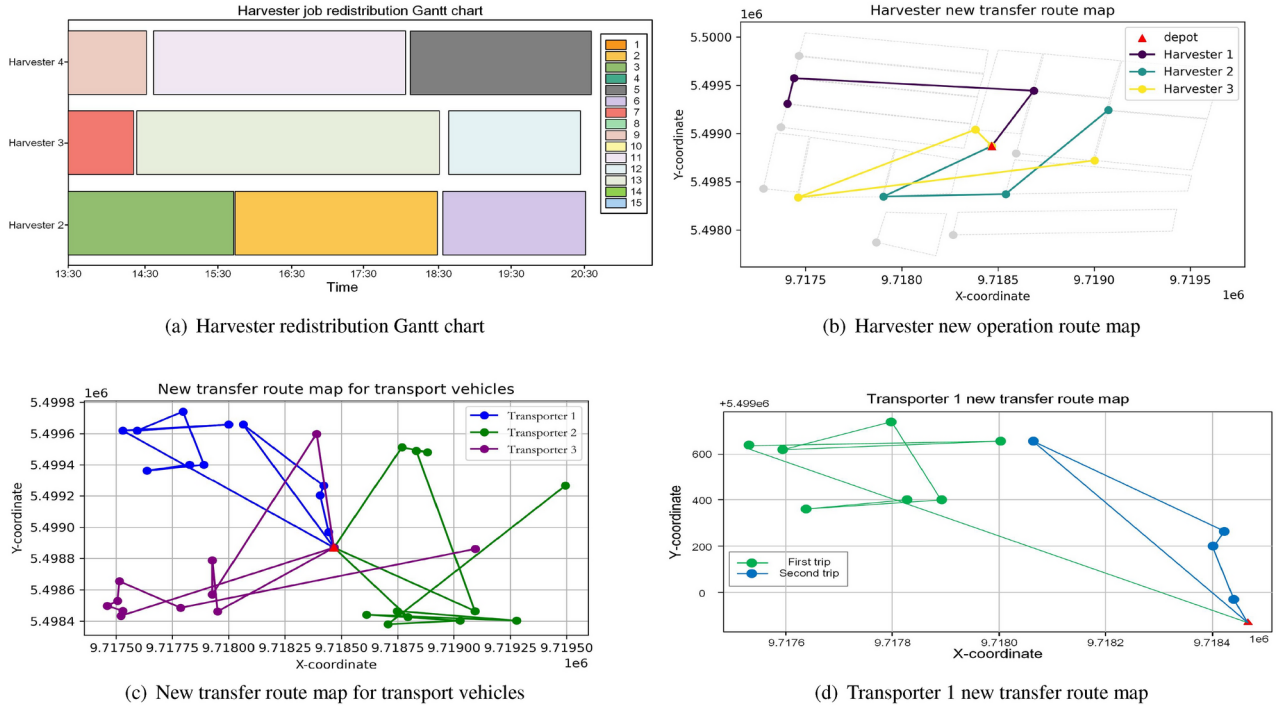
Table 4. Reallocation scheme in dynamic harvester scenarios (The farmland number in the table starts from 0 and D stands for warehouse, also in the following table).

are devised for each scenario to determine the harvester operation allocation plan minimizing objective function f_1 and the transport vehicle transportation plan minimizing objective function f_2. The reallocation scheme for harvester dynamics is detailed in Table 4. Experimental results demonstrate the algorithm's capability to correctly identify optimal solutions amidst dynamic changes in harvester numbers. Taking the second row of Table 4, where harvester 1 fails at 13:30, as an illustration, the Gantt chart for harvester job reallocation and the new transfer path are depicted in Fig. 9a,b respectively. In response to this scenario, 33 new demand points emerge, requiring a minimum of 3 transport vehicles. The optimal transport vehicle transport path and the path for transport vehicle 1 are illustrated in Fig. 9c,d.

In this experimental simulation, we investigate the following scenarios: Scenario 1: Initially consisting of 15 farmlands, the operation on farmland number 4 is canceled. Scenario 2: Starting with 15 farmlands, operations on two farmlands, numbered 4 and 10, are canceled. Scenario 3: Beginning with the first 13 farmlands numbered 0–12, a new farmland, numbered 13, is introduced to the operation. Scenario 4: Beginning with the first 13 farmlands numbered 0–12, two new farmlands, numbered 13 and 14, are introduced to the operation. Table 5 illustrates the reallocation scheme for these dynamic farmland scenarios.

During collaborative operations, the harvester and the farmland do not undergo any changes. In the event of a transport vehicle failure, to maintain the harvester's unproductive waiting time at 0, a replacement transport vehicle must be dispatched promptly to resume the tasks of the failed vehicle. Failure to do so would result in the current transport vehicle being unable to fulfill the on-time transportation plan for all demand points. Experimental simulations indicate that when one or two transport vehicles are introduced at 11:30 and 13:30, the transport vehicle redistribution plan is detailed in Table 6.

The aforementioned three sets of experiments simulated the rescheduling of harvesters and transport vehicles under various potential dynamic situations, validating the viability of the multi-machine collaborative dynamic job allocation method proposed in this article. Table 4 provides clear insights: Firstly, there is a notable upward trend in the minimum number of required transport vehicles as the number of harvesters increases. This phenomenon becomes evident when examining the distribution map of demand point timings. The



(a) Harvester redistribution Gantt chart

(b) Harvester new operation route map

(c) New transfer route map for transport vehicles

(d) Transporter 1 new transfer route map

Fig. 9. Rescheduling result graph.

Initial field number	Dynamic situation	Dynamic moment	The harvester redistributes the seqence	f_1(min)	Number of demand points	Minimum number of transport vehicles	Transporter 1 Redistributes the sequence	f_2 (km)
15	N-1	11:30	[[13, 5, 2], [3, 1, 11], [6, 12], [8, 10]]	424|18:34	41	4	[D, 0, 1, 33, 34, 35, 25, 26, D, 27, 28, 29, 30, D]	27.11
		13:30	[[11, 5], [2, 1], [6, 12], [8, 10]]	304|18:34	29	4	[D, 0, 1, 24, 25, 26, 27, 28, D, 21, D]	18.729
	N-2	11:30	[[13, 11, 5], [3, 12], [6, 2], [8, 1]]	375|17:45	36	4	[D, 0, 1, 2, 3, 4, 5, 15, D, 16, 8, D]	28.394
		13:30	[[11, 1], [2, 5], [6, 12], [8]]	304|18:34	22	4	[D, 0, 1, 9, 3, 5, 6, D]	15.726
13	Static job assignment		[[9, 10, 11, 4], [6, 5, 12], [7, 8, 1], [3, 2, 0]]	573|17:33	60	4	[D, 30, 31, 32, 33, 34, 20, 21, D, 37, 38, 39, 25, 26, 27, 28, D, 59, D]	36.498
	N+1	11:30	[[10, 11, 13], [5, 4, 1], [8, 0], [2, 12]]	429|18:39	39	4	[D, 0, 1, 2, 3, 4, 5, 6, D, 7, 36, 37, D]	28.141
		13:30	[[11, 4], [12, 13], [8, 1], [2, 0]]	243|17:33	24	4	[D, 0, 1, 2, 3, 4, 5, D]	14.687
	N+2	11:30	[[10, 11, 14], [5, 0, 1], [8, 4, 13], [2, 12]]	455|19:05	46	4	[D, 0, 1, 37, 38, 15, 16, 17, D, 30, 31, 20, 21, 22, D]	28.535
		13:30	[[11, 13, 4], [12, 0], [8, 14], [2, 1]]	354|19:24	29	4	[D, 0, 1, 2, 3, 4, 22, 23, D, 8, D]	17.059

Table 5. Reallocation scheme in farmland dynamic scenarios.

Dynamic situation	Dynamic moment	Surplus demand points	Transporter 1 Redistributes the sequence	f_2(km)
S stays the same	11:30	48	[D, 0, 4, 8, 12, 16, 20, 24, D, 28, 32, 36, 40, 44, D]	24.232
	13:30	35	[D, 0, 4, 8, 13, 16, 20, 24, D, 28, 32, D]	20.106
S+1	11:30	48	[D, 0, 7, 15, 16, 23, 27, 31, D, 35, 43, 46, D]	31.179
	13:30	35	[D, 0, 5, 10, 18, 22, 26, 30, D]	20.922
S+2	11:30	48	[D, 0, 7, 15, 16, 24, 28, 36, D, 41, D]	33.733
	13:30	35	[D, 0, 8, 13, 15, 23, 31, D]	21.427

Table 6. Transport truck redistribution plan.

rationale behind this trend is that with an augmented number of harvesters, the demand times of various points become more concentrated and distributed in a multi-segment manner. This, in turn, renders it challenging for a limited number of transport vehicles to reach all demand points within the designated transportation time period. Secondly, the experiments, based on the results of static collaborative job allocation, reveal that different dynamic scenarios occurring simultaneously lead to slight variations in the number of demand points after rescheduling. This discrepancy arises from the assumption that the harvester directly transports the last wheat back to the warehouse after its operation, excluding it from the demand points. Consequently, with an increase in the number of harvesters, there is a marginal decrease in demand points. Conversely, as demonstrated in Table 5, an increase in the number of farmlands corresponds to an upward trend in the number of demand points, while the minimum number of transport vehicles remains constant at 4. This leads to the conclusion that the minimum number of transport vehicles is positively correlated with the number of harvesters, while the number of demand points is positively correlated with the number of farmlands.

The re-scheduling of harvesters, prompted by dynamic situations in either the harvester or farmland, induces complete alterations in subsequent demand points. Therefore, in the case of double-layer rescheduling, the total distance covered by transport vehicles is not a direct point of comparison. Table 6 operates under the premise that demand points remain unchanged, focusing solely on the rescheduling of transport vehicles. The results elucidate that scheduling with the minimum number of transport vehicles yields the shortest total transportation distance. As the count of transport vehicles increases, the total distance proportionally enlarges. This serves as a validation of the efficacy of scheduling using the minimum number of transport vehicles as proposed in this paper.

Effect analysis of improved algorithms

In the algorithm section, we analyzed the limitations of applying the basic ant colony algorithm to solve the multi-machine collaborative operation allocation model. Specifically, the basic ant colony algorithm struggled to reflect the diversity of multiple agricultural machines. To address this issue, we proposed an ant colony multi-level cooperation solution. Additionally, we identified that the ant colony algorithm faced challenges in finding effective solutions for transport vehicle scheduling due to complex constraints. To overcome this, we introduced a rule-ant colony fusion algorithm.This section presents experiments designed to evaluate the performance of both the basic ant colony algorithm and the improved ant colony algorithm. For ease of analysis, the top 20 demand point entries, sorted by demand time in Table 3, are selected as algorithm inputs. Both the basic ant colony algorithm and the improved ant colony algorithm are deployed to allocate tasks to varying numbers of transport vehicles. Each experiment set is simulated 20 times to derive the optimal values. The resulting data is presented in Table 7.

From Table 7, we can draw three clear conclusions: First, the minimum number of transporters obtained by IACA is smaller than that of ACA. In the transporter scheduling module, our goal is to minimize the total distance traveled by the transporters while ensuring that the harvester's non-productive waiting time is zero and minimizing the number of transporters. Table 7 shows that ACA can only provide a feasible solution with 5 transporters, whereas IACA achieves a feasible solution with just 4 transporters. Thus, the minimum number of transporters for ACA is 5, while for IACA it is 4. Second, the optimization result of IACA surpasses that of ACA. Besides minimizing the number of transporters, our optimization objective is to reduce the total distance traveled by the transporters. We use the formula [(ACA's optimal value-IACA's optimal value)/ACA's optimal value] to express the improvement rate of IACA over ACA. When the number of transporters is 4, 5, 6, and 7, IACA's optimal values for the transporter objective function f2 are 100%, 27.3%, 0.10%, and 0.05% higher than those of ACA, respectively. Third, as the number of transport vehicles decreases, the lifting efficiency increases; conversely, as the number of transport vehicles increases, the lifting efficiency decreases. The values of 100%, 27.3%, 0.10%, and 0.05% indicate that as the number of transport vehicles increases, the improvement of IACA over ACA diminishes.

We analyze the reasons for the above three conclusions. Sorting the demand point information from Table 3 by demand time reveals that demand points exhibit small intervals within groups and large intervals between groups. This is because when the number of harvesters is 4 in the first step, the four harvesters set out to each

Number of transport vehicles	The algorithm used	Optimal transport vehicle scheduling sequence	f_2(km)	Optimization improvement rate
4	ACA	NULL	INF	100%
	IACA	[[D, 0, 4, 8, 12, 16, D], [D, 1, 5, 9, 13, 17, D], [D, 2, 6, 10, 15, 19, D], [D, 3, 7, 11, 14, 18, D]]	10.695	
5	ACA	[[D, 0, 9, 13, 18, D], [D, 1, 10, 16, D], [D, 2, 15, 17, 19, D], [D, 3, 5, 6, 8, 12, D], [D, 4, 7, 11, 14, D]]	20.522	27.30%
	IACA	[[D, 0, 8, 16, 19, D], [D, 1, 5, 9, 15, 18, D], [D, 2, 6, 10, 13, D], [D, 3, 7, 11, 14, D], [D, 4, 12, 17, D]]	14.918	
6	ACA	[[D, 0, 19, D], [D, 1, 9, 10, 16, D], [D, 2, 13, 17, D], [D, 3, 7, 11, 18, D], [D, 4, 8, 14, 15,D], [D, 5, 6, 12, D]]	19.617	0.10%
	IACA	[[D, 0, 6, 13, 19, D], [D, 1, 7, 14, 18, D], [D, 2, 10, 12, D], [D, 3, 11, 15, D], [D, 4, 8, 16, D], [D, 5, 9, 17, D]]	17.663	
7	ACA	[[D, 0, 16, 18, D], [D, 1, 10, 14, D], [D, 2, 15, 17, D], [D, 3, 7, 11, D], [D, 4, 8, 12, D], [D, 5, 9, 19, D], [D, 6, 13, D]]	19.815	0.05%
	IACA	[[D, 0, 7, 14, D], [D, 1, 9, 12, D], [D, 2, 10, 17, D], [D, 3, 11, 18, D], [D, 4, 8, 16, D], [D, 5, 13, 19, D], [D, 6, 15, D]]	18.769	

Table 7. Comparison of algorithm results.

farmland for harvesting at the same time, and the transfer time between the farmlands is not much different. Therefore, the demand points generated by the four harvesters for harvesting (unloading points when the harvester bin is full) are similar in demand time. The first demand point from each harvester forms a group, the second demand point forms another group, and so on. Thus, intervals within groups are small, while intervals between groups are large.

In ACA for transport vehicle scheduling, if a vehicle selects one of the demand points within a group, its service time exceeds the demand time of other points in the same group. Consequently, the vehicle can only select demand points from the subsequent group of time periods. If a later time point is chosen, earlier demand points cannot be served, leading to potential solution failure.

IACA improves upon this by introducing sub-ants, where each sub-ant represents a transport vehicle selecting parallel paths, enhancing search comprehensiveness and convergence speed. Additionally, a rule-based method generates a series of feasible solutions to initialize pheromones, addressing the blind search problem of ACA and mitigating round-trip oscillation. Thus, IACA can find feasible solutions more effectively than ACA, even with a stringent constraint of only four transport vehicles.

With five, six, and seven transport vehicles, each vehicle should prioritize earlier and closer demand points. For example, when the number of transport vehicles is 5, a transport vehicle can choose the first demand point within the first time period or the second time period. As the number of vehicles increases, the time constraint weakens, allowing ACA to eventually find effective solutions after multiple iterations. However, IACA still outperforms ACA. Through this experiment, it can be verified that the IACA proposed in this paper does have a certain improvement on the two problems existing in the model, and is better than ACA in minimizing the number of transport vehicles and minimizing the total distance of transport vehicles.

To gain a more comprehensive understanding of the performance differences between IACA and ACA, we analyze the scheduling results for varying numbers of demand points. However, since real farm data is difficult to obtain and unique, we generate random data sets based on real data sets. The coordinates of the demand points are random numbers uniformly distributed in the range [0,1), which are expanded by 1000 times during calculation. To closely approximate real-world data, we generate the demand point times as follows: First, the benchmark start time is set to 480 min (08:00). Second, we establish an inter-group interval u and increase it by a slightly larger number a whenever the demand point index is a multiple of the inter-group interval. Within each group, each demand point has a small random perturbation b based on the time of the previous demand point. This method aligns with the demand time distribution characteristics discussed earlier. The inter-group interval u corresponds to the number of harvesters, set to $u = 4$, with $a = 30$ and b drawn from a Uniform(0,2) distribution. We generate datasets with demand point scales ranging from 10 to 80 and schedule them using both ACA and IACA. In order to prevent the situation where the number of transport vehicles is too small and an infeasible solution occurs, we also take the number of transport vehicles as 5, based on the situation in Table 7 where the number of harvesters is 4 and ACA can only find a feasible solution when the number of transport vehicles is 5. Each group of experiments simulates 20 times to obtain the optimal value, and the experimental results are shown in Fig. 10.

The figure demonstrates that IACA consistently yields better optimal total distances compared to ACA across different scales. The optimization improvement rates at each scale are as follows: [0.54%, 21.08%, 21.33%, 25.90%, 26.15%, 28.11%, 31.44%, 36.60%]. These results indicate that as the scale of demand points increases, the improvement of IACA over ACA becomes increasingly pronounced.

Conclusions

Exploring a collaborative operation allocation method for multiple agricultural machines that can adapt to dynamic environmental changes can mitigate excessive reliance on manual experience and prevent the failure of the entire collaborative task when dynamic events occur. This paper focuses on the collaborative operation of harvesters and transport vehicles, studying the construction of collaborative scenarios and different rescheduling methods for six dynamic situations that may occur in this collaborative scenario. The common

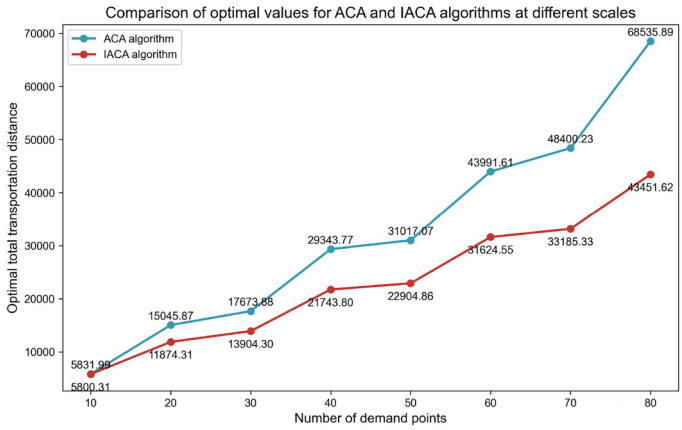


Fig. 10. Comparison of optimal values of ACA and IACA at different scales.

research goals of this study are to minimize the completion time of the latest harvester and to minimize the total distance of transporters while ensuring zero unproductive waiting time for the harvester and minimizing the number of transporters. A multi-machine collaborative dynamic job allocation model is proposed in two steps: collaborative job allocation and collaborative dynamic job allocation. And the algorithm has been adaptively improved. Using wheat harvesting and unloading in a real scenario as an example, three types of simulation experiments were conducted on the multi-machine collaborative dynamic job allocation method based on the improved ant colony algorithm. First, the feasibility of the static collaborative job allocation method was verified. Based on static job allocation, dynamic job reallocation was implemented in six scenarios. Comparative experiments were conducted between ACA and IACA. The results show that IACA solves the challenging problem of transport vehicle scheduling to a certain extent and effectively reduces the number of transport vehicles and the total distance they travel. The research lays a foundation for further solving the scheduling and management problems of multi-agricultural machinery collaborative operations in complex farmland operating environments. However, the current research is still in the algorithm simulation stage. The next step will be to conduct integrated application research on the algorithm and software platform, along with farmland experiments, to further optimize the algorithm.

Data availibility

The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request.

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Declarations

Competing interests
The authors declare no competing interests.

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